Linux

pwd print dir
ls list
cd change dir
mkdir make dir
touch creat empty file
cat Display file content
cp copy file or dir
mv Move or Rename
rm Remove
head/tail show first/last lines
grep Search content with patterns
Cut Extract columns from structured text
Sort sort lines of text
ps List Running Process
kill Send signal to process
chmod change file permission
man manual for command.

Linux | : Pipe/ Pipeline.

Internet Layers:

Application	message.	HTTP, SMTP	Proxy, Firewall
Transport	segment/Datagram	TCP, UDP	Gateway
Network	Packet	IP, ICMP	Router
Data Link	Frame	Ethernet/wi-fi	Switch, AP
Physical	Bits.	signals.	Hub, Cable.

Vim commands.

vim filename open with vim
i enter insert mode
Esc Return normal mode
:w Save/write file
:q quit vim
:wq Save and quit
:q! quit without saving
dd delete current line
u undo last change
/Pattern Search "pattern" in file
:Set number show line numbers.

UDP / TCP.

No Hand shak / Hand shake
Real-time / Sequential.
. Sock_DGRAM / . Sock_STREAM.

HTTP Methods.

GET Retrieve a Resource
POST Creeate new Resource
PUT Replace Existing Entirely
PATCH Partially update
DELETE Remove Resource.
CRUD: Create / Read / Update / Delete.

HTTP Response:

200 ok
201 Created
204 no content.
400 Bad Request
404 Not Found
405 Method not allowed
500 Internet Server error.

Rest API:

Representation State Transfer.
resources Identified by URI
and manipulated through HTTP.

Json Structure:

object as key values.: String as value.

Python.

Dictionary: {key: value}.
Tuple: (val1, val2)

Publish API.

Curl: command to Request.
Publish & Subscribe:
Publisher → broker → Subscriber.
topic: how info are grouped and Subscribed.

MQTT:

Mosquitto / Eclipse Paho.
topics + wilde Card.

MCS (Modulation & Coding Scheme). table.

Wave length equation:

$C = v \times \lambda$ $\frac{\lambda}{4}$ length $= \frac{\lambda}{4}$

$\lambda = \frac{c}{f}$.

$S(t) = A \times \sin(2\pi f t + \phi)$.

In phase/Q component I/Q Frame work:

ASK: Amplitude shift keying
Fsk: Frequency shift keying
PSK: phase shift keying.

Advanced Modulation:

QAM: varies Amplitud + phase
MPSK: M different phase angles / Symbols.
BPSK: Binary phase. PSK.
QPSK: 4- PSK.

Medium access control.

MAC: mac address Every Device: FF:FF:FF:FF:FF:FF.

Simple Pass Loss model:

$Pr_{dBM}(d) = P_{t\,dBM} + K_{dB} - \eta \times \log_{10}(\frac{d}{d_0})$.

Antenna:

Dipole / monopole Antenna
monopole wave length:

Units.

dBm: decible milliwatts.

$P_{dBm} = 10 \cdot \log_{10}\left(\frac{P_m\ W}{1\ mW}\right)$.

bps: bits per second.

Bps: Bytes per second.

1 Byte = 8 bits.

Data Rates:

1 Mbps

Contension/Random Scheduling

ALOHA, CSMA/CA

ALOHA Collision prob: transmit p.

no-transmitte $(P-1)$

Collision: p^2

Safe: $(P-1)^2 + 2(P-1)p$.

n-devices: $\frac{1}{e} = 36.8\%$ Fundamental limit

CSMA collision

listen then send.

Scheduled/Coordinated.

TDMA, FDMA, CDMA.

TDMA: time division

A, B, C, D schedule.

Python + HTTP:

Request.get( )

Request.post( )

Bluetooth:

python + JSON.

json.load(): Parse json to Python data structure.

json.dumps(): Serialize python data to json.